KG COLLEGE OF ARTS AND SCIENCE
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KGiSL Campus, Saravanampatti, Coimbatore - 641 035

REACT JS PROJECT REPORT ON

SPORTS SLOT BOOKING SYSTEM

Submitted by

VEL PRASATH GM

24BCS257

Under the Guidance of

GOWRI

Technical Trainer

Technical Department, KG College of Arts and Science

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SPORTS SLOT BOOKING SYSTEM – MERN Stack Project

Detailed Project Documentation and Implementation Report

A comprehensive overview of a modern online sports facility management platform designed for booking and managing sports slots, tracking user reservations, and ensuring secure access to booking services utilizing the MERN software stack. The system enables users to view available time slots, book or cancel reservations, and

monitor their booking history through an interactive and user-friendly interface. It also provides administrative controls for managing slot availability and scheduling. Built with React for a dynamic frontend, Node.js and Express for a robust backend, and MongoDB for efficient data storage, the application ensures scalability, performance, and secure authentication using JWT. This platform enhances the efficiency of sports facility management by reducing manual processes and delivering a seamless digital booking experience.

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1. Abstract

The Sports Slot Booking System is a web-based application developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js) to manage and simplify sports facility bookings. It allows users to view available slots, book or cancel reservations, and track their booking history.

The system uses React for the frontend interface, Node.js and Express for backend processing, and MongoDB for data storage. JWT authentication is implemented to ensure secure login and protected access. An admin module is included to manage slot availability and scheduling.

This system reduces manual work, improves efficiency, and provides a user-friendly and secure platform for sports slot management.

2. Project Overview

The Sports Slot Booking System is a MERN stack web application designed to provide an easy and efficient way to book sports facilities. Users can view available slots, book or cancel reservations, and check their booking history through a simple interface.
The system includes an admin panel to manage slot availability and scheduling. It uses React for the frontend, Node.js and Express for backend operations, and MongoDB for storing data. JWT authentication ensures secure user access.

Overall, the project aims to reduce manual booking efforts and provide a fast, reliable, and user-friendly booking experience.

3. Problem Statement

In many sports facilities, the booking process is handled manually or through inefficient systems, leading to issues such as double bookings, lack of real-time slot availability, and poor record management. Users often face difficulty in checking available slots and managing their bookings, while administrators struggle to organize schedules effectively.

Therefore, there is a need for a digital solution that allows users to easily view, book, and manage sports slots while enabling administrators to efficiently control slot availability and reduce errors.

4. Objectives

- To develop a web-based system for easy booking of sports slots
- To provide real-time viewing of available slots
- To allow users to book and cancel reservations
- To maintain booking history for users
- To implement secure authentication using JWT
- To enable admin control for managing slot availability
- To reduce manual work and improve efficiency in booking management

5. Syllabus Mapping

This project has been deliberately structured to encompass core modern web development concepts. The following table illustrates the alignment between required syllabus topics and their practical implementation within the application.

Syllabus Topic	Project Implementation
React Hooks & State Management	Used *useState* and *useEffect* to handle dynamic UI elements like slot display, booking forms, and real-time updates.
Context API / Redux	Used for managing global state such as user login status and booking information across different components.

Node.js & Express Backend	Built RESTful APIs to handle slot viewing, booking, cancellation, and fetching booking history.
MongoDB Database Design	Designed database using Mongoose with collections for Users, Slots, and Bookings.
JWT Authentication	Implemented secure login system and protected routes using JWT tokens and middleware verification.

6. Core Features

The Sports Slot Booking System includes the following key features:

- **View Available Slots:** Users can check real-time availability of sports slots for different activities.
- **Book Slot:** Users can select and reserve through the system.
- **Cancel Slot:** Users can cancel their bookings, and the slot becomes available again.
- **Booking History:** Users can view their past and current bookings in one place.
- **Admin Slot Management:** Admin can add, update, or delete slots and manage availability.
- **User Authentication:** Secure login and registration using JWT to protect user data and access.

7 .Technology Stack

The Sports Slot Booking System is developed using the MERN stack, which includes:

- **Frontend:** React.js is used to build an interactive and dynamic user interface.
- **Backend:** Node.js and Express.js are used to handle server-side logic and API development.

- **Database:** MongoDB is used to store user data, slot details, and booking information.
- **Authentication:** JWT (JSON Web Token) is used to provide secure login and protect routes.
- **State Management:** Context API / Redux is used to manage global application state.

8. System Architecture and Working Overview

The Sports Slot Booking System follows a client-server architecture using the MERN stack. The frontend is developed using React.js, which provides a responsive and interactive user interface. The backend is built with Node.js and Express.js, which handle API requests and business logic. MongoDB is used as the database to store user, slot, and booking data.

When a user interacts with the system, the frontend sends requests to the backend through RESTful APIs. The backend processes the request, communicates with the database, and sends the response back to the frontend. JWT authentication is used to verify users and protect secure routes.

Overall, the system ensures smooth communication between frontend, backend, and database, providing a fast, secure, and efficient booking experience.

9. Functional Modules

- **User Module:** Handles user registration, login, and authentication using JWT.
- **Slot Viewing Module:** Allows users to view available sports slots in real time.
- **Booking Module:** Enables users to book slots and store booking details.
- **Cancellation Module:** Allows users to cancel booked slots and update availability.
- **Booking History Module:** Displays past and current bookings of the user.
- **Admin Module:** Allows administrators to add, update, and delete slots and manage overall availability.

10.API Design

The backend of the Sports Slot Booking System is built using RESTful APIs to handle communication between the frontend and database.

Method	Endpoint	Description
GET	/slots	Fetch all available sports slots.
POST	/slots/book	Book a selected slot.
DELETE	/slots/:id	Cancel a booked slot using slot ID.
GET	/bookings/:userId	Retrieve booking history of a specific user.
POST	/auth/register	Register a new user.
POST	/auth/login	Authenticate user and generate JWT token

11. Database Design Overview

The Sports Slot Booking System uses MongoDB as the database to store and manage all application data in a structured and scalable manner. The database is designed using Mongoose schemas to define relationships between different entities.

The main collections used in the system are **Users, Slots, and Bookings**. The **User collection** stores user details such as name, email, password, and role (user/admin). The **Slot collection** contains information about available sports slots including sport type, time, date, and availability status. The **Booking collection** stores booking details by linking users with selected slots.

Each booking record is associated with a specific user and slot, ensuring proper tracking of reservations. This relational structure within a NoSQL database allows efficient data retrieval and management, supporting real-time slot booking and updates.

12. User Workflow

The Sports Slot Booking System follows a simple and structured workflow to ensure smooth user interaction. First, the user registers an account and logs in using secure JWT authentication. After successful login, the user is directed to the dashboard where all available sports slots are displayed in real time.

The user can then select a preferred slot based on sport type, date, and time. Once a slot is selected, the booking request is sent to the backend, where it is processed and stored in the database. The system then updates the slot availability accordingly.

Users can also view their booking history to track all past and current reservations. If needed, they can cancel a booking, which immediately updates slot availability. This workflow ensures a seamless and efficient booking experience from login to slot management.

13. Testing and Validation

The Sports Slot Booking System was thoroughly tested to ensure its functionality, performance, and reliability. Different types of testing were performed to validate both frontend and backend components.

Functional testing was conducted to verify core features such as user registration, login, slot viewing, booking, cancellation, and booking history. API testing was done to ensure that all RESTful endpoints return correct responses and handle errors properly. Database testing was performed to confirm accurate storage and retrieval of user, slot, and booking data in MongoDB.

Validation checks were also implemented to ensure proper input handling, such as preventing invalid bookings and unauthorized access. JWT authentication was tested to confirm secure access to protected routes. Overall, the system performed as expected and provided a stable and secure booking experience.

14. Advantages and Expected Outcomes

The Sports Slot Booking System provides several advantages that improve the efficiency and usability of managing sports facility reservations. It replaces traditional manual booking methods with a fast, automated, and user-friendly digital system. Users can easily view real-time slot availability, book or cancel slots, and track their booking history without any confusion.

The system ensures secure access through JWT authentication, preventing unauthorized usage and protecting user data. It also reduces administrative workload by allowing automated slot management and updates.

The expected outcome of this project is a fully functional and scalable booking platform that improves time management, reduces booking conflicts, and enhances the overall user experience for both users and administrators.

15. Deployment and Submission Requirements

The Sports Slot Booking System is deployed as a full-stack web application using the MERN stack. The frontend (React.js) is built and hosted separately, while the backend (Node.js and Express.js) is deployed on a server environment. MongoDB Atlas is used for cloud database storage to ensure secure and scalable data management.

For deployment, environment variables such as database connection strings and JWT secret keys are configured properly to maintain security. The application is tested after deployment to ensure all APIs, authentication, and UI features work correctly in the live environment.

- **GitHub Repository:** A well-structured repository containing both client and server codebases, complete with a git ignore file to exclude environment variables and node modules.
- **Deployment Link:** The application can be hosted on reliable cloud platforms (e.g., Vercel for the React frontend, and Render for the Node.js backend).
- **README:** Comprehensive documentation within the repository outlining setup instructions, environment variable requirements, available scripts, and architectural decisions.

16. Future Enhancements

The Sports Slot Booking System can be further improved by adding advanced features to enhance usability and scalability. One possible enhancement is the integration of real-time notifications using email or SMS to inform users about booking confirmations, cancellations, and reminders.

The system can also be upgraded with payment gateway integration to allow users to pay for slot bookings online. A rating and review system can be added for sports facilities to improve service quality.

Additionally, implementing a mobile application version using React Native would improve accessibility. Advanced analytics dashboards can be introduced for admins to track bookings, peak usage times, and user activity. These enhancements will make the system more efficient, user-friendly, and suitable for real-world large-scale deployment.

17. Conclusion

The Sports Slot Booking System is a fully functional web application developed using the MERN stack to simplify and automate the process of booking sports facilities. It provides users with an easy way to view available slots, make bookings, cancel reservations, and track their booking history in a secure and efficient manner.

The system successfully integrates React.js for a responsive user interface, Node.js and Express.js for backend services, and MongoDB for data management. JWT authentication ensures secure access and protects user data.

Overall, this project improves the traditional booking process by reducing manual effort, minimizing errors, and providing a fast and reliable digital solution for sports slot management.

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